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RESEARCH-ARTICLE

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Seeds Classification and Quality Testing Using Deep Learning and YOLO v5

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ABSTRACT

Segregation of seeds of different crops grown in the mixed cropping is a major cause of concern for the farmers as well as the food industry. Also, the classification and packaging of seeds based on their quality is a challenging task for farmers and agro-industries. Moreover, Post-thrashing separation of seeds by the traditional techniques such as sieving, hand-picking, etc. is a time-consuming and tedious task. Thus, there is a need to automate seed segregation. The potential of deep learning and machine learning techniques in object detection, classification, and pattern recognition motivated the researchers to employ these techniques for the automatic segregation of seeds at the harvesting site. The techniques proposed so far focus on the classification of seeds of different crops. Limited research work is observed that focuses on the classification of seeds of crops grown as a part of mixed cropping as well as seeds of different quality standards. Also, there is a huge scope to improve the classification performance of the proposed models. The purpose of this research to develop the deep learning-based system 'Mixed Cropping Seed Classifier and Quality Tester (MCSCQT)' for accurate classification and quality testing of seeds based on their shape, color, and texture. The system is trained on the dataset comprising labelled images of healthy and diseased seeds of pearl millet and maize. It reports the highest precision and recall of 99%. The efficacy of the system in discriminating the seeds of pearl millet and maize may prove a game-changer for the food industry. Also, its capability in recognition of diseased and healthy seeds of maize enhances its utility in the food industry.

CCS CONCEPTS

•Artificial intelligence • Computer vision • Computer vision tasks

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KEYWORDS

Seed, YOLO, Object detection, Quality testing, Classification

1 Introduction

Producing high crop yield without degrading the soil fertility is the prime objective of the agriculture industry. Moreover, there is a need for optimum utilization of resources such as manures, and water. The crop rotation and mixed cropping are followed by farmers in support of the above-stated objectives [17,32]. Mixed cropping also helps to improve the food production and economy of farmers [20,37]. The crops such as Maize and pearl millets are highly grown together as a part of mixed cropping due to their similar tolerance to the dry climate, and requirement of fewer nutrients from the soil. Also, both crops require less labor-intensive than other popular crops such as wheat, rice, and sugarcane.

Moreover, there is a similarity in the harvesting season of both crops. Thus, the farmers can harvest them together and can use for meeting the dietary requirements of human and cattle. But, the yield of both the crops together is not appropriate for the food industry and agro-industries. Also, the cost price of seeds is determined by the quality of seeds as well as the type of crop. Moreover, the quality and yield of the next generation of crops are highly dependent on the quality of seeds used to grow the crops. Thus, the commercialization of the food industry has raised the demand for the packaging of seeds of different crops into different packets. Moreover, there is a requirement to segregate the seeds based on their quality standards.

The traditional techniques such as sieving and handpicking are human resource-intensive and time-consuming. Also, these techniques require expertise in seed identification and quality detection. Therefore, it is hard to filter out seeds of different varieties of crops grown as a part of mixed cropping. Also, it is difficult to segregate the damaged and poor quality seeds from the good quality seeds. Thus, there is a high demand for automating the seed classification at the harvesting zone.

The proven significance of Deep Learning (DL) and Machine Learning (ML) techniques in yield prediction [46], enhancement of contrast of images[3,40] , disease classification[27], etc. motivated the various research groups [2,5,39,44,54,55,6,7,10,14,15,22,28,35] to extend the applicability of these techniques in seed classification based on the type, size, and their quality. The authors in [24] applied SVM for the classification of healthy and defected corn seeds [49]. The SVM classifier gave an accuracy of 95.6% for healthy and 80.6% for defected seeds. The misclassification of approximately 19% was observed during the detection of defected or damaged seeds.

Continuing the same line of research, the authors in references [53], [30], [34] employed Support Vector Machine (SVM), K-means, and Deep Convolutional Neural Network (DCNN) based models for the classification of four varieties of corn seeds. They reported the training accuracy of 100% by applying the DCNN based model. But, a large number of misclassifications are observed to one category of corn while evaluating the performance of the model on the testing dataset.

Further, the authors in [33] utilized the potential of ML and DL models for automatic inspection of maize kernels. They applied k-means clustering for separation of Kernels. They also employed various models viz. ResNet, VGGNet and AlexNet for segregation of defective and non-defective kernels. The ResNet model reported the accuracy of 98.2% and outperformed the VGGNet and AlexNe models. Similarly, the authors in [20] classified the healthy and defective maize seeds using AlexNet, VGG-19, GoogleNet and SVM. Among all these models, the GoogleNet outperformed with an accuracy of 95%.

Thus, it is inferred from the above discussion that significant research works are available for the classification of healthy and defective seeds of maize. But, the segregation of seeds of maize and pearl millet are grown together as a part of mixed cropping is still underexplored. Also, limited research works are available to classify the seeds based on the crop type as well as seed quality. Also, there is a huge scope to improve the performance of the proposed models. The strong need for automating the seed segregation at the harvesting zone and unavailability of a robust and reliable system for seed segregation based on their quality as well as crop type motivated us to develop the DL-based 'Mixed Cropping Seed Classifier and Quality Tester (MCSCQT)'. The system is effective in accurate classification and quality testing of seeds based on their shape, color, and texture.

The major objectives of this manuscript are as follows:

- Preparation of imagery dataset by using seeds of pearl millet and maize. The defective and healthy seeds of maize are labelled by the expert.
- Presenting a comparative analysis of various deep learning and object detection models employed for seed detection, identification, and classification.
- Developing "Mixed Cropping Seed Classifier and Quality Tester (MCSCQT)" system by integrating DCNN and object detection and Region-Based Convolutional Neural Networks (RCNNs).

2 Literature Review

In this section, we present the details of the research works available for DL and ML-based seed classification. We highlight the advantages and lacunae of the available approaches.

The applications of hyperspectral imaging, DL, and ML have streamlined and strengthened the automation in seed industry. DL-based algorithms are applied for seed identification, classification, clustering, and quality testing. The following studies proposed by various authors [2,5,39,44,54,55,6,7,10,14,15,22,28,35] make effective use of ML and DL techniques for seed classification and testing. They employed Convolutional Neural Network (CNN) classifiers for identifying defective seeds from non-defective, detecting seed coating, separating haploid and diploid seeds, and classifying common maize seed from silage seed used for feeding animals.

The authors in reference [28] used DL models for the identification of sunflower seeds. They overcome the problem of overfitting by using optimization algorithms. The authors claimed that the optimized GoogleNet model achieved an accuracy of 95%. But, the model requires human intervention for arranging the seeds rather than keeping them in a voluminous lot. Also, the authors considered only one view of seeds for training the model. So, there is a scope to improve the robustness and reliability of the model by training it on multiple views of seeds.

To address the challenges identified in the work proposed in reference[28], the authors in [54] considered the full surface of soybean seed. They followed circumrotating mechanism for full surface detection and reported an accuracy of 98.87%. They improved the classification accuracy by employing the MobileNet model on the dataset comprising defected seeds.

Further, the authors in [55] proposed the method for the identification of Soybean seeds. They applied Pretrained CNN models viz. AlexNet, ResNet18, Xception, Inception-v3, DenseNet201 and NASNetLarge to showcase the impact of transfer learning. The authors claimed that among all models NASNetLarge reported the highest accuracy of 97.2%. The authors also claimed that integrating hyperspectral imaging with transfer learning gives higher accuracy in lower computation costs.

For extending the application of ML models in weed identification, the authors in [14] applied the naïve Bayes algorithm[31] for the identification of weed seeds based on morphological and textural features of seeds. The model reported an accuracy of 98% on the grey scale and black and white images. But, a significant decrease in the accuracy is observed for colored images.

Further, the authors in reference[2] proposed a system that uses Angular Moment (SAM), Contrast (CONT), Entropy (ENT), Variance (VAR), Inverse Moment, and Correlation (CoR) for recognition of seeds based on the texture and shapes of seeds. The system achieved a classification accuracy of 89.25% using K-means[13].

Now, the authors in [24] developed an image capturing machine to control the uncertainty in the light intensity, reflection, and shadow appearing on seeds. They employed Support Vector Machine (SVM)[49] for seed classification and reported an accuracy of 95.6% for normal and 80.6% for defected seeds. The misclassification of about 19% is a limitation in its use for real-life purposes. Following a similar line of research, the authors in references [53], [30], [34] used hyperspectral imaging for image

acquisition. They proposed a DCNN model for the classification of four varieties of corn seeds using hyperspectral imaging. They also applied SVM, and K-means for the classification of seeds. The comparison in the performance of employed models shows that DCNN based model reported the highest training accuracy of 100%. But, the system suffers from a large number of misclassifications on the testing dataset. Similar

Further, the authors in reference [39] extracted seed vigor that includes the properties of seeds. They employed the ML algorithms viz. K-Nearest Neighbor (KNN) [31], Random Forest (RF)[38] , Multi-Layer Perceptron (MLP) [1], SVM [49] and Optimum-Path Forest OPF[36] for detection of the damaged seeds. The authors claimed that OPF and SVM outperformed all the above-stated ML models and achieved an accuracy above 80%. Further, they analyzed that considering only the features extracted based on the shape reports low accuracy. Therefore, they employed DL techniques for automatic feature selection and classification of damaged seeds.

Similarly, the authors in [5] employed an MLP ML classifier and achieved an accuracy of 98.93% to classify six varieties of corn seeds. Further, the authors in [50] proposed DeepSort to classify haploid maize seeds. They applied SVM[49], Random Forest (RF)[38], and logistic regression for the classification of haploid and diploid seeds.

Now, the authors in [33] employed ResNet, VGGNet, and AlexNet models for automatic inspection of maize kernels. ResNet model outperformed these models and achieved an accuracy of 98.2%. Similarly, the authors in [20] applied AlexNet, VGG-19, GoogleNet, and SVM to classify maize seeds based on their defects. The authors claimed that GoogleNet outperformed all models with an accuracy of 95%.

To further enhance the applicability of DL techniques, the authors in references [15] [22] applied CNN classifiers for identifying the coating on maize seeds. by authors Seed Coating They used Infrared spectroscopy qualitative modeling and a softmax classifier named Stacked Auto Encoder (SAE) and SVM for detection of seed coating with the recognition rate of 90%.

Next, the authors in [7] used near-infrared hyperspectral imaging for classifying common maize and silage seeds. Both types of seeds are slightly different in appearance but considerably vary in internal structure and characteristics. They applied SVM and radial basis function neural network (RBFNN) for classification. RBFNN outperformed in all cases in identifying common or silage seeds of maize. But, the authors reported accuracy of only 88.41% for the classification of 8 varieties of maize seeds.

Now, the authors in [6] worked for the identification of haploid and diploid seeds using CNN and transfer learning. The authors used CNN classifiers AlexNet[4], VGGNet[45], GoogeleNet[12], and ResNet[16] models for the identification of haploid and diploid seeds based on the color characteristics of their embryos. The haploid seeds have colorless embryos whereas diploid seeds have red-purple-colored embryos. The VGG-19 model outperformed all the models employed in [6].

Apart from the identification of defective seeds, various researchers [10,35,48] focused on testing the quality, purity, and genuineness of seeds. The authors in reference [35] achieved an accuracy of 90.11% using the DCNN technique. Similarly, the Seed quality tester developed by [10] divides the maize seeds into five clusters on the basis of their quality labelled as average, bad, excellent, good, and worst. The model achieved an accuracy of 81% on the training dataset. But, the model could not segregate a voluminous lot of bad seeds from good and worst categories.

To further improve the quality testing, the authors in [18] used low-latency image recognition. The system localizes the seeds and then classifies them using the ResNet-18 classifier. The system achieved an accuracy of 99.58%. The system is also effective in the segregation of weeds from crop seeds. But, some crop seeds are also segregated while removing weed seeds.

This extensive study of literature on seed classification shows that significant works are available for the identification of seeds, and quality testing but limited works are available that focus on quality testing as well as classification of maize and pearl millet, grown together as a part of mixed cropping. Also, the DL-based approaches proposed in literature overcome the drawbacks of traditional approaches of seed identification based on their texture and color.

Simultaneously, we also explored the role of object detection techniques in seed detection, disease detection, etc. We observed that object detection is a key ability required by most computer vision systems. As per the latest research in this area, the integration of object detection with deep learning models has great significance in the agriculture sector for disease detection and fruit counting.

After studying the related literature, the authors found the clue that integration of DCNN with object detection techniques 'Region-Based Convolutional Neural Network' (RCNN) may prove significant for seed quality testing. This can be further understood by the following research works.

The authors in reference [52] experimented with VGG16[45], ResNet50, and ResNet101[16] models combined with faster R-CNN for the detection of tomato disease. They claimed that the ResNet101 outperformed and reported the mAP of 90.87% on the tomato dataset comprising 207 images.

Further, the authors in reference [29] used the improved YOLO V3 model for object detection. The model inherited the features of darknet 53. Based on the comparison in the performance of YOLO V3, faster R-CNN, SSD, and YOLO V3 methods, the authors claimed that YOLO V3 improves the accuracy of object detection as it uses multiscale feature fusion on the image pyramid. It also saves computation time. The authors claimed that YOLO achieved the highest accuracy of 92.395% on the dataset of 15,000 images of tomatoes labeled as healthy and diseased.

Another group of researchers explored the integration of the DCNN and Single Shot Detector (SSD) approach. They used the concatenation of GoogleNet, Inception[47] module, and Rainbow models. The authors trained their system on the dataset consists of 26,377 images captured from the Apple Experiment Station of Northwest A&F University in Baishui County, Shaanxi province, China [23]. Their approach provides the real-time detection of apple leaf diseases with an mAP of 78.80%. Further, the authors in

reference [19] applied the MobileNet model with faster R-CNN and SSD on the dog's dataset[8]. They claimed that the MobileNet provides the real-time detection of the object with minimum latency. At the same time, the authors also stated that the models such as VGG, ResNet, and AlexNet increase the computation time.

Based on the above discussion, it is clear that the Faster R-CNN, SSD, and YOLO are suitable for object detection. Also, it is concluded that the 'YOLO' and SSD models are efficient in identifying and classifying the objects from the complete image [25,26], [23]. Also, it has a faster processing speed than other models listed above. On the other hand, the RCNN model uses the region-based approach for object detection and Yolo unifies all the separate components of the object detection model. In this approach, the network does not look at the complete image. So, the detection is done as a pipeline that is optimized in the end-to-end network [41].

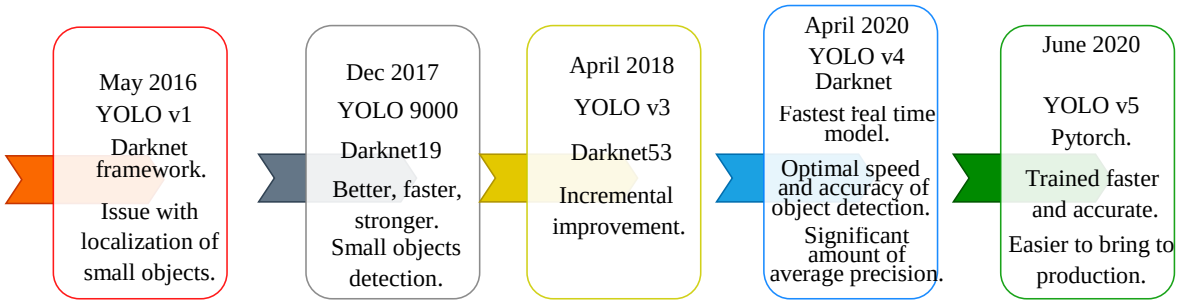

Figure 1: Evolution of YOLO object detection model.

Based on the performance analysis, it is observed that the YOLO model has poor performance in the detection of small objects. Therefore, the improved versions of YOLO such as YOLOv4 and YOLO v5 are introduced. YOLOv4 is the object detection model with optimal speed and accuracy. Similarly, the latest version YOLOv5 is the fastest implemented model in PyTorch [11,41–43]. YOLO v5 is 88% smaller than YOLO v4. Evolution of different versions of YOLO shown in figure 1.

After an extensive survey of the research works available in object detection, we chose YOLO v5 for the research work proposed in this manuscript.

4 Materials and Methods

In this section, the authors describe the architecture and experimental results of the" Mixed Cropping Seed Classifier and Quality Tester" (MCSCQT) as shown in figure 2.

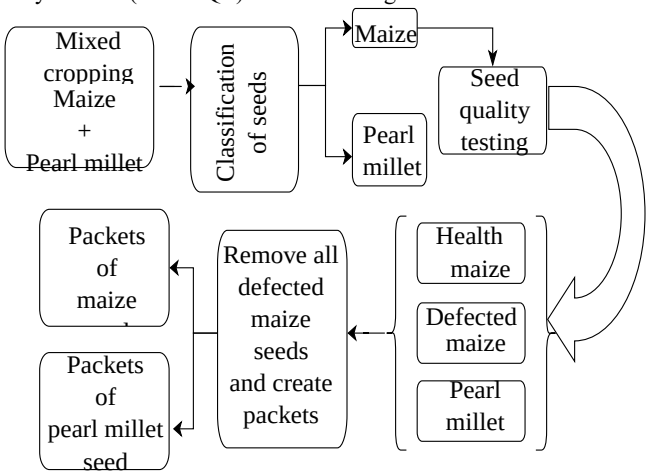

Figure 2: Architecture of Mixed Cropping Seed Classifier and Quality Tester (MCSCQT).

4.1 Image acquisition

The authors selected the seeds of maize and pearl millet grown in the mixed cropping and harvested together. The RGB images of healthy maize and pearl millet were acquired using a NIKON D750 camera, using shutter speed 1/160 sec., focal length f/6.3 (50mm), and ISO-640. Images captured on a white sheet with constant lighting conditions. The images, with 24-bit color depth and resolution equal to 2264 X 1520 pixels were stored in JPG format. The authors collected a dataset comprising 1,806 images labelled as maize_excellent, 1,500 maize_bad, 504 pearl millet, and 144 clustered images. The number of training, testing, and validation images used are 3162, 396, and 396 respectively. The number of images captured for each class is shown in Table 1.

We also used the open-source dataset available at the reference [9]. This dataset contains defective seeds of maize crops. The sample images of collected seeds are shown in figure 3.

Table 1: Dataset used for training of MCSCQT

Class	Maize_ Excellent	Maize_ Bad	Peral millet	Clustered (pearl millet + maize	Total number of images
Number of images	1,806	1500	504	144	3954

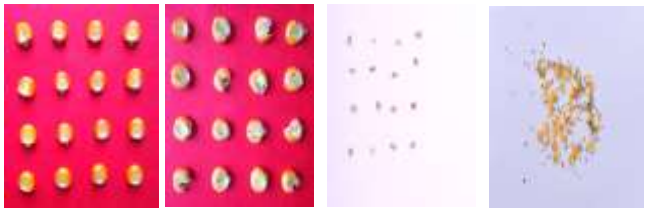

Figure 3: Sample images of the Dataset used for training MCSCQT

4.2 Pre-processing

The images captured vary in the number of pixels, shape, contrast, color, and brightness. Therefore, it is important to bring these images to a unified format. Also, the size of the dataset is small for training the DL model. Moreover, the dataset contains images of one view of seeds. Thus, it does not ensure the reliability of the model in a real-time scenario. To improve the robustness and reliability of the proposed model in a real-time scenario, we applied data augmentation techniques such as cropping, flipping, and rotation to the training datasets. After applying these augmentation techniques, the size of the dataset increased by 3 folds. Now, the dataset is converted to grayscale, which added 3 more folds. Hence the augmented dataset is 6 times the original dataset of 659 images. Half were in colored form and others in grayscale for YOLO v5[21] implementation. We applied a k-means clustering algorithm on the augmented dataset images dataset to segregate samples into maize excellent, maize bad, and pearl millet as shown in figure 4.

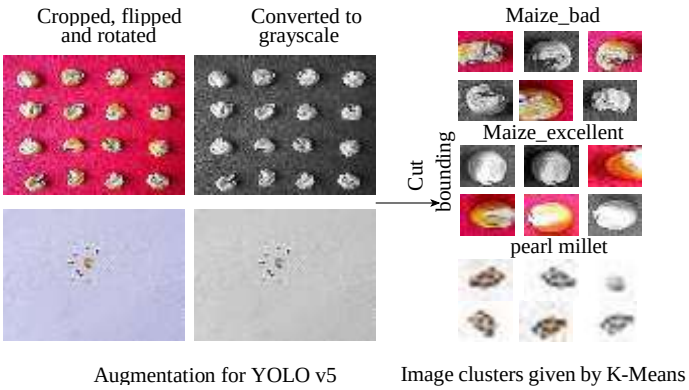

Figure 4: Dataset augmentation for YOLO v5 and k-means

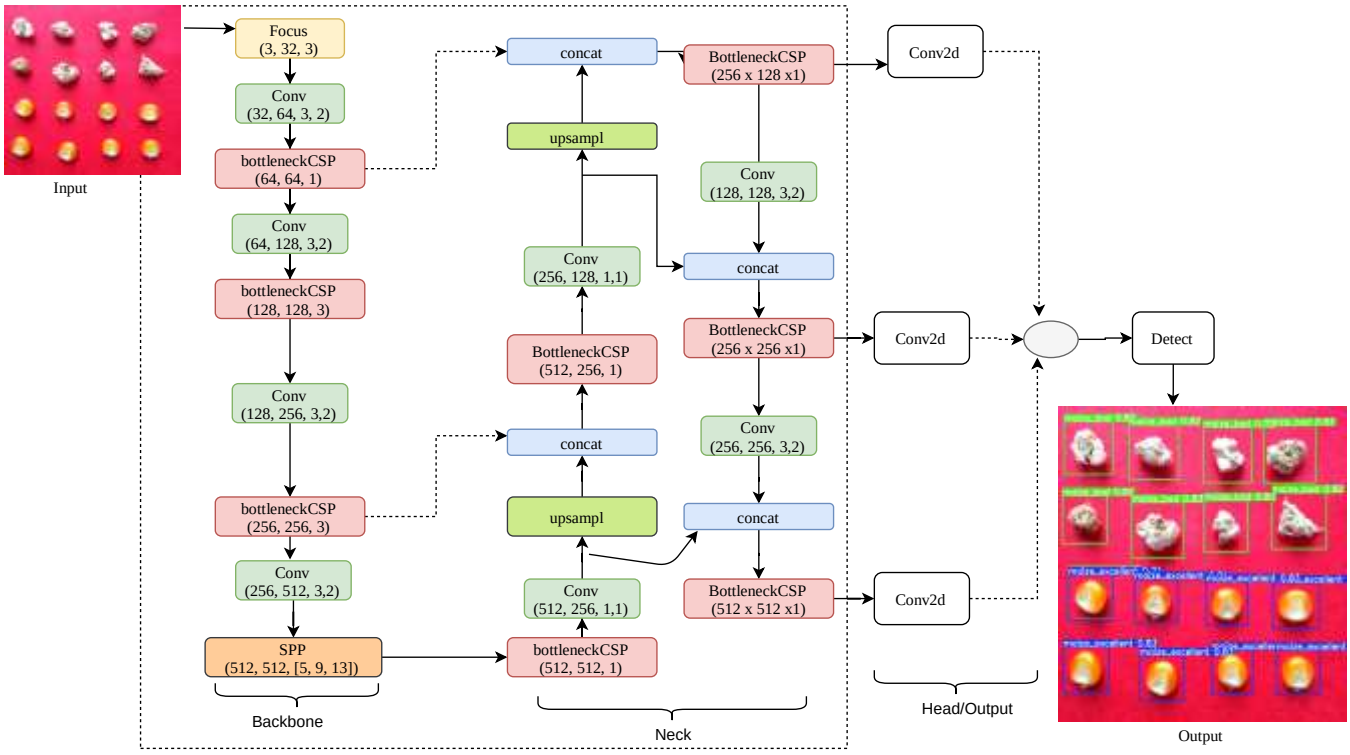

Figure 5: YOLO v5 model architecture

4.3 Model architecture

In this section, the authors demonstrate the YOLO v5 model architecture used to predict classes labelled as Maize_excellent, Maize_bad, pearl millet, and clustered [pearl millet + maize] seeds.

The architecture of "YOLO v5" is shown in figure 5. It comprises three parts- backbone, neck, and head. Model backbone is used to extract features from the maize seed image. The Cross Stage Partial (CSP)[51] is used as a backbone to extract features from a seed image in low processing time. Now, the neck part of the model is used to generate pyramids of extracted features. Pyramids are necessary for unseen data. In the YOLO v5 model, Path Aggregation Network (PANet) is used as its neck. The head or output part of the YOLO v5 model is used to perform final detection in seed image by applying anchor boxes on features. The main contribution of YOLO v5 is making training procedures fast by implementing them in Pytorch instead of Darknet. Also, the data loader in YOLO v5 augments data online and can detect even small objects with high accuracy and efficacy.

4.4 Training Resources and Hyperparameters

The authors used the resources available at the google collab for implementing the proposed "Mixed Cropping Seed Classifier and Quality Tester" (MCSCQT). The server provides 12 GB RAM, NVIDIA Tesla GPU for up to 1.5hrs experimentation. Based on the experiments conducted in [21], the authors employed momentum of 0.937, the initial learning rate of 0.01, and a final learning rate of 0.2 for the batch size of 16. An experiment was performed on 300 epochs completed in 1.5hrs.

4.5 Experimental Results

In this section, the authors present the experimental results obtained on employing YOLO v5 and k-means on a mixed cropping seed dataset. To measure the performance of 'Mixed Cropping Seed Classifier and Quality Tester (MCSCQT)' using YOLO v5, we used the evaluation metrics viz. precision, recall and mAP at various levels. It is evident from figure 6 that YOLO v5 gives the highest precision, recall, and mAP of 99%, 99%, and 98.3% respectively. Whereas, K-means algorithm. gives the precision and recall of 79.56% and 79.94% respectively. Thus, it is clear from the figure that YOLO v5 gives higher values of precision and recall than K-Means.

Further, bounding box annotation results to detect maize_bad, maize_excellent, and pearl millet seeds are shown in figure 7. It is observed that all the 3,162images used for training are correctly labeled by the model.

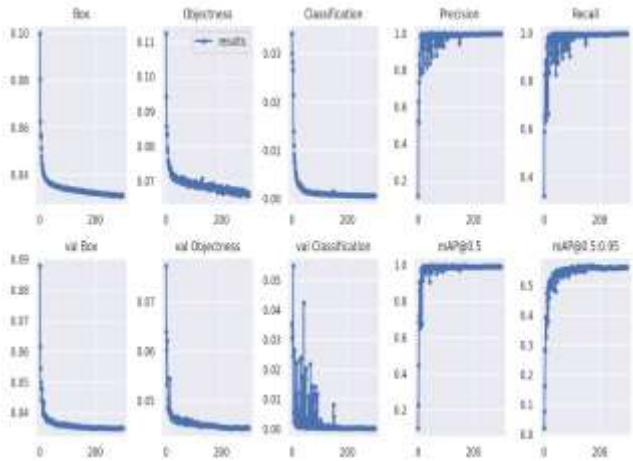

Figure 6: Precision, recall and mAP@0.5 for YOLO v5

Figure 7: Detection using YOLO v5

5 Conclusion

In this manuscript, the deep learning-based 'Mixed Cropping Seed Classifier and Quality Tester (MCSCQT)' system is proposed for automating the seed classification and quality testing. The YOLO v5 algorithm is employed with the MCSCQT for object detection. For training the system, the real dataset comprising 3,954 images of seeds of pearl millet and maize is prepared. The dataset contains 1,806 images of maize_healthy seeds, 1,500 maize_diseased seeds, 504 pearl millet, and 144 clustered images of both pearl millet and maize. The dataset size is further increased by employing the image augmentation techniques viz. flipping and rotation. After training, the system achieved the precision and recall of 99% in classifying the seeds into two classes viz. pearl millet and maize. Further, the system is found effective in segregating the diseased and healthy seeds of maize. Thus, the system may prove a game-changer for farmers, seed packaging, and the food industry by classifying the seeds of different crops grown together as a part of mixed cropping. The system is also useful for automating the packaging of seeds based on their quality. Therefore, it is an important assisting tool for increasing profit in the agricultural industry by minimizing human intervention and saving time. In the future, there is a scope to improve the potential of DL and object detection algorithms in

the detection of seeds from the voluminous heap collected at the harvesting zone.

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